

BLUEPRINTIQ TECHNICAL MEMORANDUM

UNCERTAINTY-AWARE BLUEPRINT-TO-BOQ INTELLIGENCE | DETAILED ANALYSIS REPORT

STATUTORY ENGINEERING NOTICE: BlueprintIQ provides automated preliminary computer-vision geometry extraction and material estimations. This document does not replace a licensed professional engineer, architect, quantity surveyor, or municipal authority approval. All quantities must be field-verified before procurement.

1. PROJECT & DRAWING METADATA

Project Name:	villa 3	Building Type:	Commercial
Floors:	2	Unit System:	METRIC
Target Area:	180.0 m²	Soil Type:	Not Provided
Drawing Scale:	Unknown	Concrete Grade:	M20
Wall Thickness:	230 mm	Overall Confidence:	MEDIUM
Overall Confidence Score:	87% (Estimated AI/CV Analysis)	Estimation Basis:	Computer-vision floor-plan extraction & quantity modeling

2. CONSOLIDATED MATERIAL ESTIMATES (UNCERTAINTY-AWARE)

MATERIAL	ESTIMATED	RANGE	CONF.	CALCULATION BASIS
Cement (OPC/PPC 50kg Bags)	645.0 bags	594 - 710	MEDIUM	Concrete (557.4 bags) + Masonry (36.3 bags) + Plaster (51.5 bags)
River Sand / M-Sand	45.8 m³	41 - 51	MEDIUM	Concrete (29.0m³) + Masonry (7.6m³) + Plaster (9.2m³)
Coarse Aggregate (20mm & 10mm Graded)	58.7 m³	53 - 65	MEDIUM	RCC Concrete (58.1m³) + PCC Bed (0.7m³)
Modular Clay / Fly Ash Bricks	10,567.0 nos	9,721 - 11,624	HIGH	20.13 m³ net masonry x 500 bricks/m³ + 5% breakage allowance
Reinforcement Steel (Fe500 TMT Rebars)	5,172.0 kg	4,396 - 6,464	MEDIUM	Slabs (3391.0 kg at 0.8% vol) + Beams/Columns (1780.0 kg at 1.5% vol)
Vitrified Floor Tiles (600x600mm)	159.4 m²	151 - 175	MEDIUM	Net floor area 0m² + 8% cutting and breakage allowance
Emulsion Paint & Primer (2 Coats)	90.0 litres	81 - 104	MEDIUM	Total plaster & ceiling area 900.0m² / coverage 10 m²/Litre

3. DETAILED BILL OF QUANTITIES (CIVIL & FINISHES)

CODE	DESCRIPTION	QTY	UNIT	RANGE	CONF.
CIV-01	Earthwork excavation in foundation trenches	11.14	m³	7.8 - 16.2	LOW
CIV-02	Plain Cement Concrete (PCC 1:4:8) in foundation bed (100mm)	0.81	m³	0.7 - 0.9	MEDIUM
CIV-03	Reinforced Cement Concrete (M20) for Slabs, Beams, Columns	69.12	m³	63.6 - 77.4	MEDIUM
CIV-04	Brick Masonry in Cement Mortar (1:6) with Modular Bricks	20.13	m³	18.1 - 22.1	HIGH
CIV-05	Cement Plastering (12mm Internal 1:6 + 18mm External 1:4)	540.00	m²	502.2 - 594.0	MEDIUM
CIV-06	Flooring with Vitrified Tiles including 8% wastage	159.41	m²	151.4 - 172.2	MEDIUM

4. BLUEPRINT ANOMALIES & CODE COMPLIANCE FINDINGS

ID	SEV.	FINDING & EVIDENCE	RECOMMENDED ACTION
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ISSUE-001	HIGH	Drawing Scale Not Confirmed or Uncalibrated Drawing scale specified as 'Unknown' without dimension calibration points. <i>Standard: IS 1200 / ISO 128 Architectural Drawing Presentation Rules</i>	Perform scale calibration against a known reference dimension on the drawing before procurement. Status: Verification Required
ISSUE-002	HIGH	Sub-Standard Door Opening Width (Door (91%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-003	HIGH	Sub-Standard Door Opening Width (Door (86%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-004	HIGH	Sub-Standard Door Opening Width (Door (78%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-005	HIGH	Sub-Standard Door Opening Width (Door (73%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-006	HIGH	Sub-Standard Door Opening Width (Door (63%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-007	HIGH	Sub-Standard Door Opening Width (Door (58%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-008	HIGH	Sub-Standard Door Opening Width (Door (29%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required
ISSUE-009	HIGH	Sub-Standard Door Opening Width (Door (21%)) Detected door span: 0.60 m <i>Standard: IBC 2021 Section 1010.1.1 / NBC 2016 Part 3</i>	Verify rough opening (RO) size to ensure ADA / egress compliance. Status: Verification Required

5. LOCAL AI (GEMMA 2B) SYNTHESIS & REASONING

Executive Synthesis: Preliminary architectural evaluation performed for Commercial project. Detected 9 potential dimensional or compliance considerations (9 require urgent review). Soil parameter was 'Not Provided', requiring geotechnical confirmation before finalizing foundation estimates.

Uncertainty Assessment: Overall analysis confidence is rated as MEDIUM. Confidence is constrained by 2D plan annotations and missing structural section schedules.

6. MISSING PARAMETERS & ENGINEERING ASSUMPTIONS

Missing Input Parameters:

- **Soil Type:** Not Provided — *Foundation trench excavation, footing depth, and sub-base quantities cannot be verified without soil bearing capacity (SBC).*
- **Scale Calibration:** Uncalibrated — *Drawing scale relies on nominal text ratio ('1:100') rather than two-point physical dimension calibration. Coordinate errors may magnify calculated areas by 5-10%.*

Applied Engineering Assumptions:

- Excavation depth assumed at nominal 1.2m for medium dense soil; footing sizes subject to soil report.
- Graphic scale ratio Unknown assumed uniformly proportional across X and Y axes.
- Reinforcement steel estimated via standard IS 456 thumb rules (0.8% for slabs, 1.5% for framed elements).